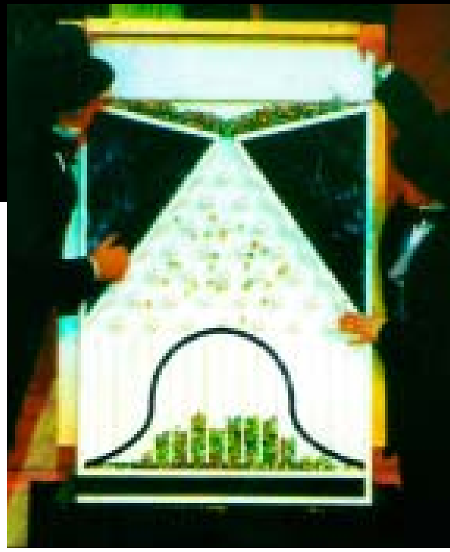
Chinese mathematicians knew about Pascal's triangle at least 900 years ago



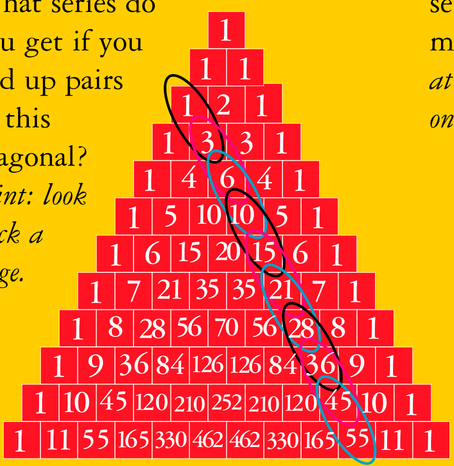
Pascal's pinball

Pascal's triangle has links to two very important branches of maths: probability and statistics. You can see why with a device called a Galton board, where marbles are poured through a pinball table with nails arranged like Pascal's triangle. The probability of a marble ending up in a particular column is easy to work out by looking at the numbers in Pascal's triangle. The final pattern is a shape called a bell curve – the most important graph in statistics.

Where are the *patterns*?

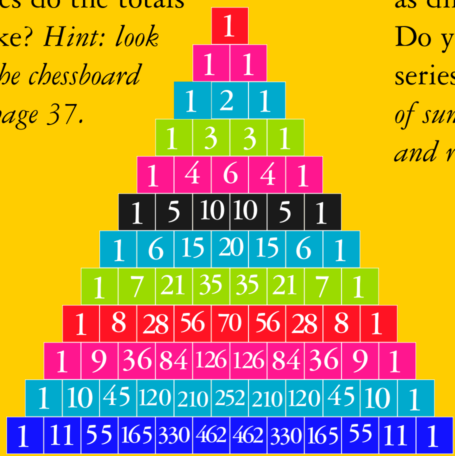
Pascal's triangle is full of fascinating number patterns. The most obvious one is in the second diagonal row on each side – the series of whole numbers. See if you can recognize the patterns below.

What number series is in the third diagonal? What series do you get if you add up pairs in this diagonal? *Hint: look back a page.*



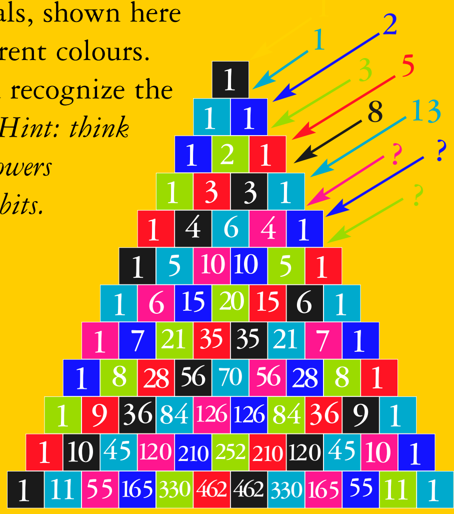
Answer: the triangular and square numbers

Add up the numbers in each row. What series do the totals make? *Hint: look at the chessboard on page 37.*



Answer: powers of two

Add up the shallow diagonals, shown here as different colours. Do you recognize the series? *Hint: think of sunflowers and rabbits.*



Answer: the Fibonacci series

The road from A to B

Here's a puzzle you can solve with Pascal's triangle. Suppose you're a taxi driver and want to drive from A to B in the town on the right. How many routes are possible? To help solve the puzzle, count the routes to nearby junctions and fill in the numbers.